(PV-SV-04) Software Verification Plan

Document ID: `PV-SV-04`
Product: `Portview`
Document Status: `Released`

Document Approval

Prepared by

Title	Name	Signature
Manager	`S. R. Lim`	
Staff	`J. B. Kim`	
General Manager	`S. I. Choi`	

Reviewed by

Title	Name	Signature
Manager	`M. C. Boo`	

Approved by

Title	Name	Signature
CTO (Director)	`K. Y. Ro`	

Revision History

Rev.	Date	Description
`0.0`	`2012.03.07`	Initial Version

Rev.	Date	Description
`0.1`	`2015.01.12`	User Interface Update
`0.2`	`2016.01.19`	New Function Implemented
`0.3`	`2017.01.13`	System Issue
`0.4`	`2018.01.12`	New Device Added
`0.5`	`2019.01.21`	Device Compatibility
`0.6`	`2020.01.30`	Device Compatibility
`0.7`	`2020.10.08`	GUI update
`0.8`	`2021.02.26`	Linkage usability improve
`0.9`	`2021.09.10`	Program Issue
`1.0`	`2021.10.08`	Device Compatibility
`1.1`	`2022.03.04`	Connection status improve
`1.2`	`2023.01.04`	Increasing image capacity
`1.3`	`2023.04.21`	Device Compatibility
`1.4`	`2024.01.08`	Device Compatibility
`1.5`	`2024.01.29`	Document number changed from 603 to Z01 according to OP-709
`1.6`	`2025.07.24`	Revision numbers of related software validation documents updated

1. Purpose

This document defines the software verification plan used to demonstrate that the applicable Portview software functions satisfy their verification and validation criteria.

The plan is intended to:

- define the verification approach for the covered product baseline
- identify the inputs, procedures, and environments used for verification
- define the acceptance logic for unit, integration, and system verification evidence

2. Scope

This plan covers the verification effort for the Portview software set.

In scope:

- software requirement verification for Portview
- interface and integration verification for the Portview software interfaces referenced in the RS inputs
- execution of the planned unit, integration, and system-level procedures

The current plan identifies software version `2.2.5.16` and the default Genoray testing platform as the principal baseline references. Software version `2.2.5.16` is associated with the `2024-10-02` release update for Windows 11 and GenX-CR compatibility.

3. Referenced Documents

The following references govern this plan for the current release baseline.

Reference	Use
Medical Device Regulation (EU) 2017/745	Regulatory framework
Quality management systems - Requirements	Quality-system reference
EN ISO 13485:2016 Medical devices Quality management systems	Governing quality management standard
ISO 12052:2006 DICOM including workflow and data management	Governing interoperability standard
IEC 62304:2006/AMD1:2015 Software lifecycle processes	Governing software lifecycle standard
Guidance for the Content of Premarket Submissions for Software Contained in Medical Devices	Regulatory guidance
General Principles of Software Validation Final Guidance for Industry and FDA Staff	Regulatory guidance
Software and Medical Devices (VdTUV, 2001)	External guidance reference

Reference	Use
QM Quality System Documentation	Internal quality reference
QM Design Control Procedure	Internal process reference
QM Software Development Procedure	Internal process reference
QM Corrective and Preventive Actions	Internal process reference
QM Document Control	Internal process reference
`PV-SV-01` Software Validation Report	Validation basis and release conclusion
`PV-SV-02` Software Development Planning	Lifecycle and configuration-management context
`PV-SV-03` Software High Level Design	Architecture and decomposition reference
`PV-SV-05` Software Verification Report	Companion verification result report
`PV-RS-01` RS for Portview	System-level requirement input
`PV-SRS-01` SwSRS for Portview	Software requirement input
`PV-SDS-01` SwSDS for Portview	Software design input
`PV-TM-01` Traceability Matrix	Requirement and design traceability reference
`PV-STP-01` SwSTP for Portview	Unit verification procedure
`PV-TP-01` SwTP for Portview	Integration verification procedure

4. Verification Objectives

The verification effort should demonstrate the following:

- the applicable Portview software requirements are verified by controlled procedures and recorded evidence
- the integrated behavior between Portview software components and interfaces is verified
- the planned verification procedures are defined clearly enough to support repeatable execution
- the final verification evidence can support a release or approval decision through an explicit pass or justified-fail logic

5. Verification Strategy

Verification is organized by level. The plan is structured around unit verification, integration testing, and system testing, with the procedure set serving as the primary evidence backbone.

5.1 Test Levels

The plan preserves the principal verification levels and document pairings.

Level	Goal	Primary Inputs	Primary Outputs
Unit	Verify requirement-level behavior of Portview software units	`PV-SRS-01` SwSRS for Portview	`PV-STP-01` SwSTP for Portview
Integration	Verify integrated behavior against RS-level interface requirements	`PV-RS-01` RS for Portview	`PV-TP-01` SwTP for Portview
System	Verify system-level behavior on the controlled product baseline	System-level specifications and integrated configuration	`SystemTP-Z01`

5.2 Verification Methods

Planned methods include:

- code review before unit-level verification
- execution of controlled unit-level procedures against the designated Portview requirement inputs
- integration testing against RS-derived interface requirements
- system testing using `SystemTP-Z01`
- review of execution results and final release decision through the verification report

5.3 Acceptance Criteria

All procedures referenced by this plan shall be executed, and the final status of each required result shall be either:

- `Passed`, or
- `Justified` when a failure or deviation is accepted through formal review and documented disposition

The final verification report should not claim successful completion unless this rule is met.

6. Verification Environment

The currently available information identifies assigned personnel and the core execution platform. Additional configuration detail can be appended if it is maintained as a separate controlled record.

6.1 Hardware And Devices

Item	Identifier	Revision	Notes
Product under test	`Portview`	`Software baseline 2.2.5.16`	Release dated `2024-10-02`; Windows 11 and GenX-CR compatibility update
System test platform	`Genoray testing platform`	`Configuration: Default`	System procedure identifier is `SystemTP-Z01`
Integration interfaces	`Portview configuration`	`Serial/prototype number: -`	Hardware and network details remain to be expanded if controlled separately

6.2 Software And Tools

Tool	Version	Purpose	Validation Status
`PV-SRS-01` SwSRS for Portview	`Rev 1.6`	Unit-level requirement input	`Released`
`PV-STP-01` SwSTP for Portview	`Rev 0.1`	Unit-level test procedure	`Released`
`PV-RS-01` RS for Portview	`Rev 0.1`	Integration requirement input	`Released`
`PV-TP-01` SwTP for Portview	`Rev 0.1`	Integration test procedure	`Released`
`PV-SYSTP-01` SystemTP-Z01	`Rev 1.6`	System-level test procedure	`Released`

Tool	Version	Purpose	Validation Status
Microsoft Visual C++ 2017	`Professional Edition`	Build environment	`Validated per PV-SV-02 Section 7.3`
Genoray testing platform	`Default configuration`	Test execution environment	`Validated per PV-SV-02 Section 7.3`

6.3 Roles And Responsibilities

Role	Responsibility	Owner
Software Team Manager	Verification oversight for Portview software activities	`S.I. Choi`
Software Engineer	Execution and support for software verification tasks	`S.R. Lim`
Software Engineer	Execution and support for software verification tasks	`J.W. Lee`

7. Verification Items

The verification items below follow the planning structure used in this document and keep the verification scope aligned with the defined procedure set.

Item ID	Description	Source Requirement / Design Input	Verification Level
`UNIT-PORTVIEW`	Portview unit verification	`PV-SRS-01` SwSRS for Portview / All Sections	`Unit`
`INT-PORTVIEW`	Portview integration verification	`PV-RS-01` RS for Portview / Section 2.1 Software Requirements	`Integration`
`SYS-PORTVIEW`	Portview system verification	`System-level integrated baseline`	`System`

The verification items above are tied to the following planned procedure set:

Verification Item	Planned Procedure	Notes
Portview unit verification	`PV-STP-01` SwSTP for Portview / All Sections	Code review is completed before this verification
Portview integration verification	`PV-TP-01` SwTP for Portview / All Sections	`PV-RS-01` RS for Portview Section 2.1 Software Requirements is used as the input
Portview system verification	`PV-SYSTP-01` SystemTP-Z01	System testing is conducted with `SystemTP-Z01`

If finer-grained requirement grouping is needed, it should be maintained in a separate traceability annex instead of replacing the planning structure in this section.

8. Procedures And Evidence

The following procedure set forms the execution baseline for this plan.

Procedure ID	Title	Evidence Output	Owner
`PV-STP-01`	SwSTP for Portview	Unit-level Portview execution records	`Software team`
`PV-TP-01`	SwTP for Portview	Integration execution records for Portview interfaces	`Software team`
`PV-SYSTP-01`	SystemTP-Z01	System-level execution records	`System verification owner`

9. Schedule And Entry/Exit Criteria

This repository does not yet contain an active execution schedule. Until that information is added, the following gate logic should apply:

- entry: approved baseline, approved procedures, available test environment, assigned personnel
- exit: required procedures executed, deviations dispositioned, verification report updated, release decision prepared

10. Deviations And Risk Handling

Any failed, blocked, or partially executed procedure shall be recorded with:

- the affected item or procedure identifier
- the observed problem
- the impact on safety, effectiveness, or release confidence
- the disposition owner
- the justification or remediation plan

Deviations should not be embedded solely in narrative text. The report set should maintain them in a dedicated table.

11. Traceability Strategy

Traceability will be maintained from requirement inputs to procedures, and from procedures to execution evidence and final summary claims.

Traceability rules:

- each procedure should name its controlling requirement or document section
- each executed record should reference the procedure identifier and baseline
- the verification report should summarize status by verification level and unresolved deviation status